

1 Environmental Parameter Calculations

1.1 Standard Atmospheric Pressure Correction

The function `pressure_at_elevation` calculates the pressure P based on the International Standard Atmosphere (ISA) model:

$$P(H) = 1013.25 \cdot (1 - 2.25577 \times 10^{-5} \cdot H)^{5.25588} \quad (1)$$

Variable Definitions & Reasoning:

- $P(H)$: Atmospheric pressure (hPa) at a specific elevation.
- H : Elevation in meters ($elevation_ft \times 0.3048$).
- 1013.25: Standard sea-level pressure (hPa).
- 2.25577×10^{-5} : Lapse rate constant derived from standard temperature decrease (0.0065 K/m) over sea-level temperature.
- 5.25588: Hydrostatic exponent determined by gravity g and the gas constant for air R .

1.2 Vapor Pressure

The saturation vapor pressure $e_s(T)$ uses the Magnus formula, while the actual vapor pressure e_a is derived using Relative Humidity (RH):

$$e_s(T) = 6.112 \cdot \exp\left(\frac{17.67 \cdot T}{T + 243.5}\right) \quad , \quad e_a = e_s(T) \cdot \frac{RH}{100} \quad (2)$$

2 Psychrometry and Wet Bulb Temperature (T_w)

The `wet_bulb_f` solves the non-linear psychrometric equation:

$$e_s(T_w) - A \cdot P \cdot (1 + 0.00115 \cdot T_w) \cdot (T - T_w) - e_a = 0 \quad (3)$$

Variable Definitions:

- $A = 6.66 \times 10^{-4}$: Psychrometric constant representing the balance between latent heat of evaporation and convective heat transfer.

3 Solar Geometry Derivation

The function `estimate_solar_position` computes the solar elevation (h) and azimuth (A_{sun}).

3.1 Orbital Phase and Declination (δ)

$$B = \frac{2\pi}{365}(N - 81) \quad , \quad \delta = 23.45^\circ \cdot \sin(B) \quad (4)$$

Variables: N is the Day of Year; 81 aligns the sine wave with the Vernal Equinox; 23.45° is the axial tilt of Earth.

3.2 Equation of Time (EoT) and Local Solar Time

$$EoT = 9.87 \sin(2B) - 7.53 \cos(B) - 1.5 \sin(B) \quad (5)$$

$$\omega = 15^\circ \cdot (UTC_{hour} + \frac{\lambda}{15} + \frac{EoT}{60} - 12.0) \quad (6)$$

Variables: ω is the solar hour angle (0 at solar noon); λ is longitude; Φ is latitude.

4 Radiative Balance in Complex Terrain

4.1 Shortwave Radiation (SW_{net})

Accounting for slope (β) and aspect (A_{slope}):

$$\cos(\theta_i) = \cos(\beta) \sin(h) + \sin(\beta) \cos(h) \cos(A_{sun} - A_{slope}) \quad (7)$$

$$SW_{net} = 1361.0 \cdot \cos(\theta_i) \cdot (1 - 0.65c^2) \cdot (1 - \alpha) \quad (8)$$

Reasoning: 1361.0 is the solar constant; 0.65 is the cloud attenuation coefficient; α is the surface albedo (default 0.7 for snow).

4.2 Longwave Radiation (LW_{net})

Using the Sky View Factor ($SVF = \cos^2(\beta/2)$):

$$\epsilon_{atm} = (1 - c)(0.51 + 0.066\sqrt{e_a}) + c \quad (9)$$

$$LW_{net} = \sigma[SVF \cdot \epsilon_{atm} T_{air}^4 + (1 - SVF) \epsilon_{terrain} T_{air}^4 - T_{snow}^4] \quad (10)$$

Reasoning: $\epsilon_{terrain} = 0.98$ for surrounding rocks/trees; $T_{snow} = 273.15K$ assumes a melting snow surface.

5 Effective Temperature Synthesis (T_{eff})

$$T_{eff} = T_{wet} + \frac{SW_{net} + LW_{net}}{K_{rad}} \quad (11)$$

Variables: $K_{rad} = 8.0$ is the radiant heat transfer coefficient in $W/(m^2 \cdot K)$.